Liquidity Score Changelog

Full version history of Liquidity Score methodology decisions, from v1.0 to v5.7.

ump to Version

Latest v5.7 v5.6 v5.5 v5.4 v5.3 v5.2 v5.1 v5.0 v4.9 v4.8 v4.7 v4.6

LATEST VERSION

v5.7 May 20, 2026

Peg-aware staged discovery price gate

Secondary discovery pools now need a plausible tracked-token price before their TVL can be staged or merged into Liquidity Score aggregates.

IMPACT SNAPSHOT

- CoinGecko Onchain and GeckoTerminal pool rows whose tracked-token price fails the existing peg-aware DEX observation sanity gate are rejected before staging
- Already-staged rows with implausible tracked-token prices are skipped during scoring merge, so malformed discovery rows cannot inflate coin or global TVL until their staging TTL expires
- Carbon DeFi chain-suffixed secondary-source ids now normalize to the DefiLlama `carbon-defi` protocol cap, adding a cap-level backstop for Carbon rows
- Rows without a measured token price still flow through the existing TVL, volume, dedupe, protocol-cap, and retained-pool quality gates

v5.7 May 20, 2026

Peg-aware staged discovery price gate

Secondary discovery pools now need a plausible tracked-token price before their TVL can be staged or merged into Liquidity Score aggregates.

- CoinGecko Onchain and GeckoTerminal pool rows whose tracked-token price fails the existing peg-
- Already-staged rows with implausible tracked-token prices are skipped during scoring merge, so

aware DEX observation sanity gate are rejected before staging
- Carbon DeFi chain-suffixed secondary-source ids now normalize to the DefiLlama `carbon-defi` protocol cap, adding a cap-level backstop for Carbon rows

malformed discovery rows cannot inflate coin or global TVL until their staging TTL expires
- Rows without a measured token price still flow through the existing TVL, volume, dedupe, protocol-cap, and retained-pool quality gates

Hide details

IMPACT NOTES

- CoinGecko Onchain and GeckoTerminal pool rows whose tracked-token price fails the existing peg-aware DEX observation sanity gate are rejected before staging
- Already-staged rows with implausible tracked-token prices are skipped during scoring merge, so malformed discovery rows cannot inflate coin or global TVL until their staging TTL expires
- Carbon DeFi chain-suffixed secondary-source ids now normalize to the DefiLlama `carbon-defi` protocol cap, adding a cap-level backstop for Carbon rows
- Rows without a measured token price still flow through the existing TVL, volume, dedupe, protocol-cap, and retained-pool quality gates

Commit provenance not recorded

v5.6 May 7, 2026

Staged discovery TVL sanity ceiling

Secondary discovery rows now reject impossible pool TVL before staging and skip any already-staged over-cap row during the scoring merge.

- CoinGecko Onchain, GeckoTerminal, DexScreener, and CoinGecko ticker staging can no longer persist non-finite, negative, or over-cap TVL values
- Valid high-liquidity pools below the ceiling continue through the existing dedupe, protocol-cap, and retained-pool quality gates
- The scoring cron skips legacy staged rows above the same sanity ceiling before they can affect global TVL, coverage drift, or DEX price observations

Details

v5.5 Apr 17, 2026

Absolute TVL Depth fallback recalibration and Slipstream sqrt_ratio price

Absolute TVL Depth fallback (used when circulatingUsd is unavailable) now shares the ratio formula's anchor via a $1B implied reference mcap. Aerodrome/Velodrome Slipstream price derivation now uses on-chain sqrt_ratio (Q64.96) instead of total-reserve ratios for concentrated liquidity pools.

- Absolute TVL Depth fallback: `20 * log10(tvl / 100_000) + 20` → `35 * log10(tvl / 700_000)`; coins without market cap data no longer gain ~24 points of unearned TVL Depth
- Aerodrome/Velodrome Slipstream price observations now derive from on-chain sqrt_ratio instead of reserve ratios; concentrated liquidity pools no longer emit biased spot prices when one side lacks a tracked USD price
- Slipstream pools where sqrt_ratio is unusable and one side has no tracked price are now dropped entirely (no reserve-ratio fallback derivation)

Details

v5.4 Apr 14, 2026

Curve enrichment scoping and staged UUID dedupe

Curve API enrichment is now scoped to Curve DeFiLlama rows, and staged exact pool-id rows can dedupe against a single identity-poor DeFiLlama UUID row through the same narrow optional-metadata wildcard used by primary dedupe.

- Non-Curve DeFiLlama pools that share token symbols with a Curve pool no longer inherit Curve registry metadata, balance ratios, token prices, or metapool-adjusted TVL
- CoinGecko/GeckoTerminal provider ids with underscores or provider suffixes normalize to the same canonical protocol family as DeFiLlama ids during pool-identity construction
- Staged discovery now skips a staged exact pool-id row when it uniquely matches one primary DeFiLlama UUID row by chain, protocol, token set, and pool-shape family, while ambiguous same-pair staged buckets still remain separate

Details

v5.3 Apr 8, 2026

PancakeSwap trailing-hour volume window

PancakeSwap V3 direct volume now sums the official `poolHourDatas.volumeUSD` buckets across a bounded trailing 24-hour window instead of treating the latest `poolDayDatas` row as if it were a rolling 24h metric.

- Intraday PancakeSwap volume no longer collapses toward zero until UTC rollover just because the current day bucket has only accumulated partial activity
- The PancakeSwap direct fetch keeps bounded batching under The Graph row cap while staying on official subgraph entities instead of adding a new historical block lookup dependency
- Fresh non-swap day buckets can no longer zero out yesterday's still-relevant trading activity, because trailing volume now comes from summed hourly swap buckets

Details

v5.2 Apr 8, 2026

Orderbook ticker contract refresh and Balancer exact-address identity

CoinGecko orderbook fallback now ignores the deprecated `trust_score` field and validates tickers by observable freshness/price/volume fields, while Balancer direct pools now use the API's exact pool address for identity instead of the 32-byte vault pool id.

- CoinGecko tickers fallback and discovery staging no longer drop every post-March-2026 ticker row just because CoinGecko now returns `trust_score = null`
- Balancer direct API pools now reserve and dedupe by the true pool address, restoring exact-id confirmation against staged discovery and overlap checks
- Orderbook fallback ticker intake now requires finite USD price/volume plus a stable exchange identifier, improving sanitization without depending on deprecated metadata

Details

v5.1 Apr 7, 2026

Authoritative protocol confirmation for staged discovery

Staged discovery rows can no longer invent new pools inside protocol families that already have a clean protocol-native direct source. When that authoritative fetch succeeds on a supported chain, staged rows must match one of its exact pool ids or they are excluded.

- GeckoTerminal, CoinGecko Onchain, and DexScreener staging rows that claim Balancer, Fluid, Raydium, Orca, Meteora, PancakeSwap, Aerodrome, or Velodrome liquidity now require authoritative exact-id confirmation when the matching direct fetch succeeded cleanly on that chain
- The guard fails open when the authoritative source is unavailable or degraded, so discovery sources still recover coverage during native-source incidents instead of hard-zeroing the row

- Liquidity cron metadata now records `stagedPoolsSkippedByAuthoritativeProtocol` separately from exact-id and derived-identity dedupe skips

Details

v5.0 Apr 5, 2026

Size-aware scoring: relative TVL depth, recalibrated volume, quality retention

All scoring dimensions are now size-independent. TVL Depth measures effective TVL relative to market cap instead of absolute dollar value. Volume Activity has a recalibrated curve with a realistic ceiling (tops out at ~32% V/T instead of ~500%). Pool Quality measures venue quality retention ratio (qualityAdjustedTvl / totalTvl, rescaled) instead of absolute quality-adjusted TVL. Weights rebalanced to 30/20/20/20/10.

- TVL Depth uses effective-TVL-to-market-cap ratio on a log scale (35 × log10(ratio / 0.0007)), with absolute fallback for coins without market cap data
- Volume Activity recalibrated: 38 × (log10(V/T) + 3) — zero line at 0.1% V/T, tops at ~32% V/T. USDC/USDT now score 86-90 instead of 52-56
- Pool Quality measures quality retention (qualityAdjustedTvl / totalTvl, rescaled from 15-80% range to 0-100). Fully size-independent
- Weights rebalanced from 35/20/22.5/15/7.5 to 30/20/20/20/10 — structural quality (Pool Quality + Durability = 40%) now matches depth + activity (50%)

Details

v4.9 Apr 3, 2026

Blocked dead Bunni DEX inputs

Explicitly blocked Bunni from liquidity scoring and DEX implied-price publication after dead-venue rows kept surfacing as retained liquidity.

- Bunni is now excluded during crawl intake and DeFiLlama pool processing instead of being treated as a live DEX venue
- Retained-pool filters and challenger snapshots ignore Bunni even if stale rows or unexpected inputs survive earlier gates
- Liquidity scores, dexPriceUsd, and downstream DEX cross-checks no longer count Bunni TVL, pool counts, or protocol medians

Details

v4.8 Apr 3, 2026

Direct-source duplicate hardening for Balancer and staged exact ids

Direct-source dedupe now reserves every authoritative exact pool id for later staged matching, and Balancer stablecoin pools can still collapse against direct Balancer coverage when DefiLlama omits the subtype in `balancer-v3` metadata.

- Sub-threshold direct API pools now still block later exact-address staged duplicates from re-entering scoring with incompatible TVL semantics
- GeckoTerminal and CoinGecko discovery rows can no longer inflate liquidity by resurrecting the same exact direct pool after the direct row was excluded from scoring
- Balancer stablecoin pools now dedupe correctly against Balancer direct API even when DefiLlama labels the row as generic `balancer-v3` without stable subtype metadata

Details

v4.7 Apr 3, 2026

Retained-pool DEX price publication

DEX implied-price publication now derives from the final retained pool surface after dedupe, caps, and scoring filters, instead of from the earlier raw observation stream.

- Pools that are skipped as duplicates or dropped by retained-pool quality filters can no longer keep influencing dex_prices
- dexPriceUsd, price_sources_json, and downstream dexPriceCheck consumers now reflect the same retained pool surface used for challenger publication and UI liquidity detail
- High-TVL discovery rows that never survive retained-pool admission can no longer manufacture near-peg DEX aggregates for depegged assets

Details

v4.6 Mar 24, 2026

Protocol-native DEX coverage expansion

Liquidity scoring now ingests Meteora DLMM, PancakeSwap V3, and Aerodrome/Velodrome Slipstream pool-state data as protocol-native direct sources, expanding primary-grade coverage across Solana, BSC, Base, and Optimism.

- Meteora DLMM pools now enter the direct-API merge path with measured TVL, volume, balances, and fee data
- PancakeSwap V3 pools now add protocol-native primary coverage across BSC and supported EVM chains through official Graph subgraphs

- Aerodrome Slipstream and Velodrome Slipstream pools now contribute pool-state TVL, balances, fees, and DEX-price observations via the on-chain Sugar view contracts
- Direct-source precedence over overlapping DeFiLlama rows now requires measured non-zero 24h volume, so Slipstream pool-state rows expand coverage without displacing stronger DL rows when volume telemetry is absent

Details

v4.5 Mar 24, 2026

Coverage recall hardening and measurement-aware confidence

DEX liquidity now paginates deeper through GeckoTerminal and CoinGecko Onchain discovery, enriches weak partial coverage instead of only zero-coverage rows, and scores coverage confidence from measured-vs-synthetic retained TVL rather than a fixed source-family ladder.

- GeckoTerminal and CoinGecko Onchain token crawls now read multiple bounded pages instead of stopping after page 1
- Fallback orderbook rows now preserve explicit synthetic/decayed/provenance flags instead of masquerading as organic USDC pools
- DexScreener and CoinGecko tickers fallback now trigger for weak partial coverage, not only coins with zero pools or no DEX price
- Coverage confidence now incorporates protocol breadth, source-family breadth, measured balance share, measured price share, and synthetic or decayed TVL share

Details

v4.4 Mar 19, 2026

Chain-aware pool identity dedupe and challenger snapshot publishing

DEX liquidity now resolves tracked tokens chain-aware, deduplicates pools with conservative identity keys instead of coarse fingerprints, collapses duplicate DEX price observations before aggregation, and publishes dedicated challenger snapshots from the full retained pool set.

- Direct API and staged/fallback pools resolve tracked assets by chain+address first, with chain-scoped symbol fallback only when unique
- Repeated sightings of the same physical pool across direct API, staged, and fallback sources now collapse before dex_prices weighting
- Cross-source pool dedupe now uses exact pool ids first and derived token-shape matches only when they are unique on both sides
- Depeg challenger inputs publish from the full retained pool set instead of the visible top-pools subset

Details

v4.3 Mar 18, 2026

Fluid DexReservesResolver balance integration

Fluid pools on Ethereum, Arbitrum, Base, and Polygon now read balances and fee detail from the official DexReservesResolver instead of staying on neutral placeholders.

- Fluid top-pool rows now populate Balance and Detail when the official DexReservesResolver is deployed on that chain
- Fluid pool quality now uses measured balance health on resolver-backed EVM chains, rather than a hardcoded neutral 1.0
- Fluid fee detail now comes from the on-chain pool config (`1% = 10_000`), normalized to basis-point badges in the UI
- BSC and Plasma Fluid pools remain on neutral-balance fallback until Fluid ships the same resolver path there

Details

v4.2 Mar 18, 2026

Measured direct-API balance health and normalized pool-detail metadata

Balancer, Raydium, and Orca direct-API pools now preserve measured balance and fee metadata through scoring instead of merging with neutral placeholders. Pool-detail fee tiers are normalized to basis points for all sources.

- Direct-API Balancer, Raydium, and Orca pools now populate top-pool balance bars and detail badges
- Measured direct-API balance ratios now feed balance-weighted aggregates, stress, and effective TVL instead of assuming 1.0
- Balancer weighted pools normalize balance health against target token weights rather than raw reserve symmetry
- Orca vault balances are normalized from raw token units before balance-health calculation

Details

v4.1 Mar 18, 2026

Direct API precedence, primary-grade coverage, and fetcher contract hardening

Direct API sources now replace overlapping DeFiLlama pools before scoring, run ahead of staged/fallback sources, and count as primary-grade coverage. Raydium and Orca contract handling was hardened against live API drift, Fluid volume normalization moved to one-sided USD volume, and Balancer intake now excludes unsupported pool types.

- Raydium lower-case poolType contract fix restores live Solana pool coverage
- Orca now paginates via cursor.next with retry/backoff and a below-threshold stop, instead of truncating after page 1
- Direct API pools are fingerprint-deduped and preferred over overlapping DeFiLlama pools
- Direct API sources merge before staged/DexScreener/CG-ticker fallbacks,

before score computation

preventing lower-confidence sources from claiming the same pool first

Details

v4.0 Mar 10, 2026

Log-scale volume, cross-chain removal, durability rebalance

Volume activity switched from linear to log-scale. Cross-chain component removed and weight redistributed to TVL depth and pool quality. Durability sub-weights rebalanced: locked liquidity removed, organic fraction reduced to 15% with sqrt curve, history-measured signals raised to 85%.

- Volume activity now uses log-scale (33.3*log10(V/T/0.005)) — median score rises from 5 to ~35
- Cross-chain component removed; TVL Depth raised to 35%, Pool Quality to 22.5%
- Durability: organic 15% (sqrt curve), TVL stability 35%, volume consistency 25%, maturity 25%
- Locked liquidity sub-component removed from durability (no reliable data source)

Details

v3.4 Mar 9, 2026

Retained-pool score recomputation and trusted staged-price hardening

Liquidity scoring now rebuilds every aggregate from the retained pool set after filtering/caps, while staged discovery preserves pool-quality metadata and stricter DEX-price trust rules.

- Filtered or TVL-capped pools can no longer keep influencing score inputs through stale aggregate fields
- HHI now uses the full retained pool set before display truncation; global 7d volume is pool-deduped
- Staged pool merge now dedups against token-pair fingerprints and preserves raw DEX metadata/quality multipliers
- DEX price observations require a consistent $50K post-confidence TVL floor across source families

Details

v3.3 Mar 9, 2026

Separated discovery pipeline with staged pool confidence decay

Discovery sources (CG Onchain, GeckoTerminal, DexScreener, CG Tickers) now run on an independent 20-minute cron with 3x more budget. Staged pools merged into scoring with freshness confidence decay and explicit defaults contract.

- Discovery cron runs independently on 20-min trigger with ~15 min budget (was 5 min shared)
- Chain-aware source routing reduces wasted API calls by skipping irrelevant chains
- Staged pools receive confidence decay: max(0.5, 1 - ageHours/48), excluded after 24h
- Tiered priority with exponential backoff prevents looping on pool-less coins

Details

v3.2 Mar 2, 2026 Reconstructed

Effective TVL symbol-fallback inflation fix

Corrected effective TVL inflation when symbol fallback matched non-Curve pools to Curve entries.

- Metapool-adjusted TVL now applies only to address-matched Curve pools
- Removes accidental score inflation from cross-pool symbol collisions
- Symbol-fallback pools keep their own TVL in effective TVL calculations

Details

v3.1 Feb 28, 2026 Reconstructed

Anti-duplication and protocol TVL cap normalization

Introduced fingerprint-based deduplication and DeFiLlama-anchored cap logic to suppress inflated secondary-source TVLs.

- CG/GT/DS pools deduped against DeFiLlama using token-pair fingerprints
- Global protocol and chain TVL totals kept consistent after cap reductions
- Secondary-source pool TVL capped and proportionally scaled by protocol-level DeFiLlama ceilings

Details

v3.0 Feb 28, 2026 Reconstructed

Coverage expansion with fallback sources

Expanded zero-pool recovery with DexScreener and CoinGecko tickers fallbacks for orderbook-heavy assets.

- DexScreener fallback adds pools for tracked coins still missing after primary crawl
- Reduces false zero-liquidity outcomes for long-tail and niche assets
- CoinGecko tickers fallback synthesizes orderbook liquidity where AMM coverage is absent

Details

v2.2 Feb 27, 2026 Reconstructed

No-pool rows moved to NR semantics

Coins without DEX pools switched from score=0 placeholders to NULL (NR) semantics.

- No-liquidity rows now persist liquidity_score as NULL instead of 0
- Downstream consumers can distinguish not-rated from genuinely low-liquidity assets
- Daily history placeholders for no-pool coins also use NULL scores

Details

v2.1 Feb 25, 2026 Reconstructed

Onchain source upgrade and locked-liquidity durability term

Primary pool discovery moved to CoinGecko Onchain with locked-liquidity data integrated into durability scoring.

- CG Onchain became primary source (with GT fallback) for richer pool metadata
- Locked liquidity added as an explicit durability sub-component
- Durability weights changed from 40/25/20/15 to 35/25/20/15/5

Details

v2.0 Feb 19, 2026 Reconstructed

Six-component v2 liquidity model

Moved from a five-component heuristic to a six-component model with effective TVL and durability decomposition.

- Weights changed from 35/25/20/10/10 to 30/20/20/15/7.5/7.5
- Durability and per-component score breakdown persisted in D1
- TVL depth switched to effective TVL, not raw TVL only

Details

v1.0 Feb 19, 2026 Reconstructed

Initial DEX liquidity score release

Launched baseline DEX liquidity scoring, API surface, and dashboard integration.

- Initial five-component composite (TVL depth, volume, pool quality, diversity, cross-chain)
- Liquidity map endpoint and page-level leaderboard shipped
- DeFiLlama-driven pool aggregation and top-pool persistence introduced

Details

WHITE PAPER Download Liquidity Score Changelog v5.7 (PDF)